Namit Anand

Curriculum Vitæ

NASA Quantum AI Lab
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Bio

I'm a Staff Scientist working in quantum information theory and quantum computation at the Quantum AI Lab at NASA Ames Research Center and KBR. I received my PhD in Physics from the University of Southern California under the supervision of Paolo Zanardi, coadvised by Todd Brun and Aaron Lauda. Before that, I received my Integrated Master of Science in Physics from the National Institute of Science Education and Research, Bhubaneswar, India.

Research interests

Application of quantum information-theoretic tools to many-body theory (specifically, information scrambling, quantum chaos, and many-body localization), quantum resource theories (specifically, quantum coherence and entanglement), open quantum systems and non-Markovian processes, quantifying the role of coherence and entanglement in quantum algorithms, error mitigation for NISQ-devices, non-universal models of quantum computation (Clifford group, matchgates, etc.), simulation of quantum optics, benchmarking quantum devices, classical shadow tomography, qudit quantum computing.

Employment

2024 – present **Staff Scientist**, *NASA Quantum AI Lab (QuAIL) and KBR*, Mountain View, CA, USA.

2022 – 2023 **Postdoctoral Researcher**, *NASA Quantum AI Lab (QuAIL) and KBR*, Mountain View, CA, USA.

Education

2017 – 2022 **Doctor of Philosophy, Physics**, *University of Southern California*, Los Angeles, CA, USA.
Advisor: Paolo Zanardi. *Co-advisors*: Todd A. Brun and Aaron Lauda.
Thesis: Quantum information-theoretic aspects of chaos, localization, and scrambling [link]

2016 – 2017 **Visiting Student Researcher, Physics**, *Harish-Chandra Research Institute*, Prayagraj, India.

2011 – 2016 **Integrated Master of Science (BSc +MSc), Physics**, *National Institute of Science Education and Research*, Bhubaneswar, India.

Publications & Preprints

1. **Certifying the quantumness of a nuclear spin qudit through its uniform precession**, *arXiv:2410.07641 (2024)*, Arjen Vaartjes *et. al*.

2. **Benchmarking the algorithmic reach of a high-Q cavity qudit**, *arXiv:2408.13317 (2024)*, Nicholas Bornman, Tanay Roy, Joshua A. Job, Namit Anand, Gabriel N. Perdue, Silvia Zorzetti, M. Sohaib Alam.
3. **Creation and manipulation of Schrödinger cat states of a nuclear spin qudit in silicon**, *arXiv:2405.15494 (2024)*, Xi Yu *et. al.*, [to be updated online].
4. **Assessing and advancing the potential of quantum computing: A NASA case study**, *Future Generation Computer Systems (2024)*, Eleanor G. Rieffel, Ata Akbari Asanjan, M. Sohaib Alam, Anand, Namit, *et. al.*.
   – Featured as an Editors' choice in November 2024
5. **General protocols for the efficient distillation of indistinguishable photons**, *arXiv:2404.14217 (2024)*, Jason Saied, Jeffrey Marshall, Namit Anand, Eleanor G. Rieffel.
6. **Dual-Unitary Classical Shadow Tomography**, *arXiv:2404.01068 (2024)*, Ahmed A. Akhtar, Namit Anand, Jeffrey Marshall, Yi-Zhuang You.
7. **Advancing Quantum Networking: Some Tools and Protocols for Ideal and Noisy Photonic Systems**, *Proc. SPIE 12911, Quantum Computing, Communication, and Simulation IV, 1291106 (2024)*, Jason Saied, Jeffrey Marshall, Namit Anand, Shon Grabbe, Eleanor G. Rieffel.
8. **Simulation of quantum optics by coherent state decomposition**, *Optica Quantum **1**(2), 78-93 (2023)*, Jeffrey Marshall and Namit Anand.
   – Featured on the cover of Optica Quantum
9. **Scrambling and operator entanglement in local non-Hermitian quantum systems**, *Phys. Rev. B **108**, 134305 (2023)*, Brian Barch*, Namit Anand*, Jeffrey Marshall, Eleanor Rieffel, Paolo Zanardi.
   – PRB Editors' Suggestion, * = equal contribution.
10. **Scrambling of Algebras in Open Quantum Systems**, *Phys. Rev. A **107**, 042217 (2023)*, Faidon Andreadakis, Namit Anand and Paolo Zanardi.
    – PRA Editors' suggestion
11. **BROTOCs and Quantum Information Scrambling at Finite Temperature**, *Quantum **6**, 746 (2022)*, Namit Anand and Paolo Zanardi.
12. **Quantum coherence as a signature of chaos**, *Phys. Rev. Research **3**, 023214 (2021)*, Namit Anand, Georgios Styliaris, Meenu Kumari, and Paolo Zanardi.
13. **Information Scrambling and Chaos in Open Quantum Systems**, *Phys. Rev. A **103**, 062214 (2021)*, Paolo Zanardi and Namit Anand.
14. **Information Scrambling over Bipartitions: Equilibration, Entropy Production, and Typicality**, *Phys. Rev. Lett. **126**, 030601 (2021)*, Georgios Styliaris, Namit Anand, and Paolo Zanardi.
15. **Quantum coherence and the localization transition**, *Phys. Rev. B **100**, 224204 (2019)*, Georgios Styliaris, Namit Anand, Lorenzo Campos Venuti, and Paolo Zanardi.
16. **Quantifying non-Markovianity: a quantum resource-theoretic approach**, *arXiv:1903.03880 (2019)*, Namit Anand and Todd A. Brun.

17. **Demonstration of fidelity improvement using dynamical decoupling with superconducting qubits**, *Phys. Rev. Lett.* ***121****, 220502 (2018)*, Bibek Pokharel, Namit Anand, Benjamin Fortman, and Daniel A. Lidar.
    – Popular summary of our work in Phys.org, ScienceDaily, and the Scientific American.
18. **Asymmetry and coherence weight of quantum states**, *Phys. Rev. A* ***97****, 032342 (2018)*, Kaifeng Bu, Namit Anand, and Uttam Singh.
19. **Coherence and Entanglement Monogamy in the Discrete Analogue of Analog Grover Search**, *arXiv:1611.04542 (2016)*, Namit Anand and Arun K. Pati.
20. **Comment on "Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics"**, *Eur. J. Phys.* ***37*** *048003 (2016)*, Namit Anand.
21. **Do quantum strategies always win?**, *Quantum Information Processing* ***14*** *(11), 4027-4038 (2015)*, Namit Anand and Colin Benjamin.

⋆ [Google Scholar] and [arXiv].

Academic Service

Referee	QIP 2023, Physical Review X, Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review B, Physical Review E, Physical Review A, Quantum Science and Technology
Conference Organizer	YQIS 2021
Scientific writing	Phys.org

Selected Seminars

1. **IBM Qiskit Seminar**, *Information-theoretic aspects of scrambling and chaos*, postponed to 2025, [YouTube link].
2. **VirginiaTech, USA**, *Information-theoretic aspects of scrambling and chaos*, Nov. 18, 2024, [link to seminar page].
3. **University of Virginia, USA**, *Information-theoretic aspects of scrambling and chaos*, Nov. 14, 2024, [link to seminar page].
4. **Max Planck Institute of Quantum Optics, Germany**, *Information-theoretic aspects of scrambling and chaos*, Oct. 29, 2024, [link to seminar page].
5. **Atominstitut TU Wien, Vienna**, *Information-theoretic aspects of scrambling and chaos*, Oct. 22, 2024.
6. **MIT Quantum Group, USA**, *Information-theoretic aspects of scrambling and chaos*, June 21, 2024, [link to seminar page].
7. **UNSW, Australia**, *Information-theoretic aspects of scrambling and chaos*, April 10, 2024, [link to seminar page].
8. **QSI Seminar at University of Technology Sydney, Australia**, *Information-theoretic aspects of scrambling and chaos*, April 9, 2024, [link to seminar page].

9. **University of Sydney, Australia**, *Information-theoretic aspects of scrambling and chaos*, April 8, 2024.
10. **IIT Madras, India**, *Information-theoretic aspects of scrambling and chaos*, Jan. 11, 2024.
11. **IMSc Chennai, India**, *Information-theoretic aspects of scrambling and chaos*, Jan. 10, 2024.
12. **ICTS Bangalore, India**, *Information-theoretic aspects of scrambling and chaos*, Jan. 8, 2024, [link to seminar page].
13. **University of Luxembourg & Donostia International Physics Center, Spain**, *Quantum information-theoretic aspects of scrambling, localization, and chaos*, June 29, 2023, Adolfo del Campo and Aurelia Chenu's group, [link to seminar page].
14. **UNSW, Australia**, *Random unitaries, quantum chaos, and all that*, May 25, 2023, Andrea Morello's group.
15. **UC San Diego, USA**, *Quantum information-theoretic aspects of scrambling, localization, and chaos*, Oct. 4, 2022, CMT Journal Club.
16. **IIT Kanpur, India**, *Quantum information-theoretic aspects of scrambling, localization, and chaos*, Aug. 26, 2022, [Young Quantum Condensed Matter (YQCM) seminar series].

Teaching and Mentorship

(Fall '17 & Spring '18) Graduate Teaching Assistant for Physics 135aL: Physics for the Life Sciences. Student Evaluation Rating: 5.0/5.0

Summer 2024 Mentoring undergraduate students for Institute for Pure & Applied Mathematics RIPS summer program at UCLA. (June 24 - August 23, 2024)

Summer 2024 Lectures on introduction to quantum information and computation to undergraduate students for Fermilab and Brookhaven Summer School Exchange Program (July 15 - 18, August 1 - 6, 2024).

Awards and Fellowships

2022 Robust Quantum Simulation (RQS) Postdoctoral Fellowship (Declined)

Summer 2021 Recipient of the USC Gold Family Fellowship.

Award: $5,000 (USD)

Summer 2018 Recipient of the Quantum computing fellowship at the Los Alamos National Laboratory.

Award: $12,200 (USD)

2017 – 2022 University of Southern California, Department of Physics & Astronomy.

Award: $27,500 (USD) annually; full financial support through teaching assistantship and one year of fellowship for completion of the Ph.D. program.

2017 Graduate fellowships declined for USC: University of Maryland; Dartmouth College; Université libre de Bruxelles; Scuola Normale Superiore di Pisa.

Summer 2015 One of the five summer fellows (internationally) for the fully funded I.I.A.S.S. Summer internship program, Italy.

Award: €3,000.

2011 – 2016 Recipient of Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship by the Department of Science and Technology, Government of India from 2011 - 2016.

Award: ₹80,000/year for 5 years.

Computational Skills

Programming Languages	C++, Python
Specialized Software	Mathematica and MATLAB
Markup Languages	LaTeX, HTML, HTML5, PHP
Software adept	Adobe Suite (Photoshop, Audition) and Game Development (UDK)
Numerical packages	*Lapack* for C++ & Python, MATLAB, Python, Julia, C++ on HPC, and CVX, CVXPY

Professional References

Paolo Zanardi

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University of Southern California
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Eleanor G. Rieffel

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Daniel A. Lidar

(*Viterbi Professor of Engineering*)
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Todd A. Brun

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